

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 items, 1 point each — 8 points)

Write the letter of the best definition on the line beside each term. **Two definitions will not be used.**

- | | | | |
|----------|---------------------------------|----------|--------------------------|
| C | 1. Greatest common factor (GCF) | G | 5. Zero product property |
| F | 2. Prime polynomial | H | 6. Double root |
| J | 3. Perfect square trinomial | D | 7. Excluded value |
| B | 4. Difference of squares | A | 8. Rational expression |

- (A) A quotient of two polynomials.
(B) A binomial of the form $a^2 - b^2$; it factors as $(a + b)(a - b)$.
(C) The largest factor shared by every term of a polynomial.
(D) A number that makes a denominator zero, so the expression is undefined there.
(E) An expression written as a product of two or more expressions.
(F) A polynomial with integer coefficients that cannot be factored any further over the integers.
(G) If a product equals zero, then at least one of its factors equals zero.
(H) A solution that appears twice because the factor producing it is squared.
(I) The exponent that tells how many times a base is used as a factor.
(J) A trinomial of the form $a^2 \pm 2ab + b^2$; it factors as $(a \pm b)^2$.

Part B — Multiple Choice (6 items, 2 points each — 12 points)

Circle the best answer.

1. Which expression is equal to $(2m + 3)^2$?
(A) $4m^2 + 9$
(B) $4m^2 + 6m + 9$
(C) $4m^2 + 12m + 9$ ← correct
(D) $2m^2 + 12m + 9$
2. What is the value of $\sqrt{9 + 16}$?
(A) 7
(B) 5 ← correct
(C) 25
(D) 12
3. Which is the **complete** factorization of $6x^2 + 9x$?
(A) $x(6x + 9)$
(B) $3(2x^2 + 3x)$
(C) $3x(2x + 3)$ ← correct
(D) $3x(2x^2 + 3)$

4. Which polynomial is **prime** over the integers?
- (A) $x^2 - 49$
 (B) $x^2 + 49$ ← **correct**
 (C) $x^2 - 14x + 49$
 (D) $x^2 + 14x + 49$
5. How many solutions does $x^3 + 9x = 0$ have, and of what type?
- (A) 3 solutions: 3 real
 (B) 3 solutions: 1 real and 2 imaginary ← **correct**
 (C) 1 solution: 1 real
 (D) 2 solutions: 2 real
6. Which simplification is **correct**?
- (A) $\frac{x+5}{x+7} = \frac{5}{7}$
 (B) $\frac{x+5}{5} = x$
 (C) $\frac{5(x+5)}{5(x+7)} = \frac{x+5}{x+7}$ ← **correct**
 (D) $\frac{x^2+5}{x^2} = 5$

Part C — Short Answer & Computation (10 items, 5 points each — 50 points)

Show all work. An answer with no supporting work earns no credit.

1. Simplify completely. Write your answer with positive exponents only. $\frac{(3x^2y^3)^2 \cdot 2xy}{6x^3y^4}$

$$\begin{aligned}\frac{(3x^2y^3)^2 \cdot 2xy}{6x^3y^4} &= \frac{9x^4y^6 \cdot 2xy}{6x^3y^4} \\ &= \frac{18x^5y^7}{6x^3y^4} \\ &= 3x^2y^3\end{aligned}$$

2. (a) Simplify $\sqrt{75x^3}$, where $x > 0$. (b) Write $\sqrt[3]{x^5}$ using a rational exponent. (c) Evaluate $16^{3/4}$.

$$(a) \quad \sqrt{75x^3} = \sqrt{25x^2} \cdot \sqrt{3x} = 5x\sqrt{3x}$$

$$(b) \quad \sqrt[3]{x^5} = x^{5/3}$$

$$(c) \quad 16^{3/4} = \left(\sqrt[4]{16}\right)^3 = 2^3 = 8$$

3. Factor completely by grouping: $3x^3 + 12x^2 + 2x + 8$

$$\begin{aligned}3x^3 + 12x^2 + 2x + 8 &= 3x^2(x + 4) + 2(x + 4) \\ &= (x + 4)(3x^2 + 2)\end{aligned}$$

4. Factor completely: $4x^2 + 11x + 6$

$$\begin{aligned}ac &= 4 \cdot 6 = 24, \text{ and } 3 + 8 = 11 \\ 4x^2 + 11x + 6 &= 4x^2 + 8x + 3x + 6 \\ &= 4x(x + 2) + 3(x + 2) \\ &= (x + 2)(4x + 3)\end{aligned}$$

5. Factor completely. (a) $49x^2 - 64$ (b) $x^3 - 27$

$$(a) \quad 49x^2 - 64 = (7x)^2 - 8^2 = (7x + 8)(7x - 8)$$

$$(b) \quad x^3 - 27 = x^3 - 3^3 = (x - 3)(x^2 + 3x + 9)$$

6. Solve by factoring: $2x^2 = 5x + 12$

$$2x^2 - 5x - 12 = 0$$

$$(x - 4)(2x + 3) = 0$$

$$x - 4 = 0 \quad \text{or} \quad 2x + 3 = 0$$

$$x = 4 \quad \text{or} \quad x = -\frac{3}{2}$$

7. Simplify, and state **every** excluded value. $\frac{x^2 - 16}{x^2 - x - 12}$

$$\begin{aligned} \frac{x^2 - 16}{x^2 - x - 12} &= \frac{(x + 4)(x - 4)}{(x - 4)(x + 3)} \\ &= \frac{x + 4}{x + 3} \quad (x \neq 4, x \neq -3) \end{aligned}$$

8. Divide and simplify, and state **every** excluded value. $\frac{x^2 - 4}{x^2 + 6x + 9} \div \frac{x + 2}{x + 3}$

$$\frac{(x + 2)(x - 2)}{(x + 3)^2} \cdot \frac{x + 3}{x + 2} = \frac{x - 2}{x + 3} \quad (x \neq -3, x \neq -2)$$

9. A rectangular banner has area $A(x) = 2x^2 + 11x + 12$ square feet and width $x + 4$ feet. (a) Find an expression for its length. (b) Find the length, width, and area when $x = 5$.

$$2x^2 + 11x + 12 = (x + 4)(2x + 3)$$

$$(a) \quad \text{length} = 2x + 3 \text{ feet}$$

$$(b) \text{ at } x = 5: \quad \text{width} = 9 \text{ ft}, \quad \text{length} = 13 \text{ ft}$$

$$\text{area} = 9 \cdot 13 = 117 \text{ square feet}$$

10. Factor completely. (a) $5x^3 - 45x$ (b) $2x^4 - 32$

$$(a) \quad 5x^3 - 45x = 5x(x^2 - 9) = 5x(x + 3)(x - 3)$$

$$\begin{aligned} (b) \quad 2x^4 - 32 &= 2(x^4 - 16) \\ &= 2(x^2 + 4)(x^2 - 4) \\ &= 2(x^2 + 4)(x + 2)(x - 2) \end{aligned}$$

Part D — Extended Response (2 items, 15 points each — 30 points)

1. **Verify, then judge a claim.**

- (a) Verify the identity $(x - 4)(x^2 + 4x + 16) = x^3 - 64$ by multiplying.

$$\begin{aligned} (x - 4)(x^2 + 4x + 16) &= x^3 + 4x^2 + 16x - 4x^2 - 16x - 64 \\ &= x^3 - 64 \end{aligned}$$

- (b) A student writes: " $\frac{x^3 - 64}{x - 4}$ simplifies to $x^2 + 4x + 16$, for every value of x ." Test both expressions at $x = 5$ and at $x = 4$.

$$\text{at } x = 5 : \quad \frac{125 - 64}{1} = 61, \quad 25 + 20 + 16 = 61$$

$$\text{at } x = 4 : \quad \frac{64 - 64}{0} = \frac{0}{0} \text{ --- undefined}$$

$$\text{but } 16 + 16 + 16 = 48$$

- (c) Is the student's algebra correct? Is the student's *claim* correct? Explain in one or two sentences, using the word **excluded**.

Yes — the algebra is right; part (a) proves the two expressions match......

No — the claim is wrong: $x = 4$ is an excluded value of the original......

At $x = 4$ the original is $0/0$, but the simplified form returns 48......

2. **A rocket is launched from a platform.** Its height in feet after t seconds is $h(t) = -16t^2 + 64t + 80$.

- (a) Factor $h(t)$ completely and find both values of t for which $h(t) = 0$.

$$h(t) = -16t^2 + 64t + 80$$

$$= -16(t^2 - 4t - 5)$$

$$= -16(t - 5)(t + 1)$$

$$t = 5 \quad \text{or} \quad t = -1$$

- (b) Which of those two values describes when the rocket lands? What does the other value mean here, and why is it rejected?

$t = 5$: the rocket lands 5 seconds after launch, when its height is 0......

$t = -1$ is one second before launch, so it is rejected — time starts at 0......

- (c) Solve $h(t) = 80$. Interpret **both** solutions — this time neither one is rejected.

$$-16t^2 + 64t + 80 = 80$$

$$-16t^2 + 64t = 0$$

$$-16t(t - 4) = 0$$

$$t = 0 \quad \text{or} \quad t = 4$$

$t = 0$ is the launch: the rocket starts at 80 feet, on the platform......

$t = 4$ is on the way down, passing 80 feet again — both are kept......

- (d) How many solutions does the equation $h(t) = 0$ have, and of what type? How does the degree of $h(t)$ tell you this before you factor anything?

Two solutions, both real: $t = 5$ and $t = -1$

h has degree 2, and the degree gives the number of solutions......